EG9x
Reference Design

LTE Module Series

Rev. EG9x_Reference_Design_Rev.A

Date: 2018-02-09

Status: Released

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About the Document

History

Revision	Date	Author	Description
A	2018-02-09	Felix YIN	Initial

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1 Reference Design

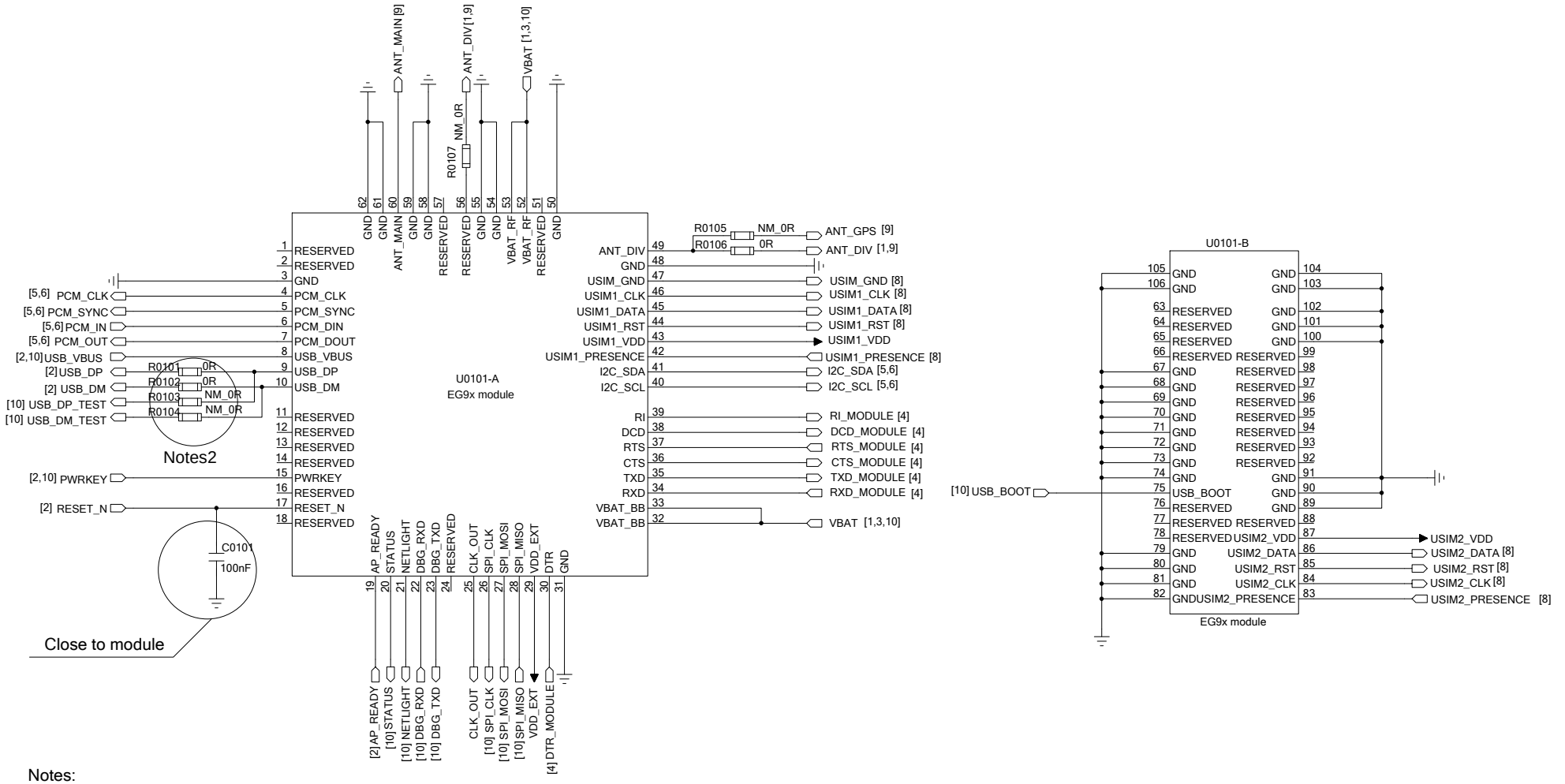
1.1. Introduction

This document provides the reference design for Quectel EG9x modules, which contain EG91 and EG95 modules.

1.2. Schematics

The schematics illustrated in the following pages are provided for your reference only.

Module Interface



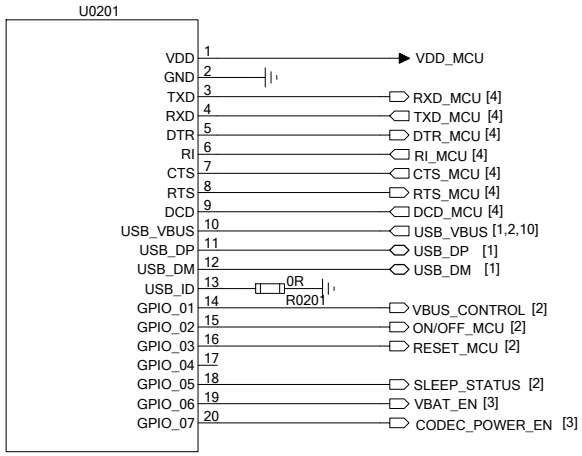
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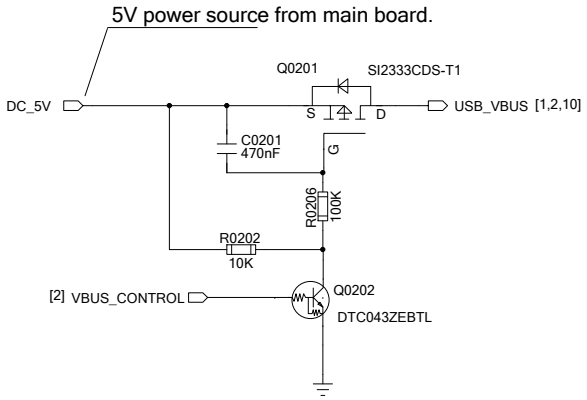
1. Keep all RESERVED and unused pins unconnected.
2. It is recommended to reserve the test points for upgrading the firmware over USB interface and minimizing stub length of USB test signals.
3. EG9x module provides a digital clock output (CLK_OUT) for an external audio codec, please refer to EG91 and EG95 Hardware Designs for further details.
4. SMD difference between EG9x-E and EG9x-NA is shown as below:

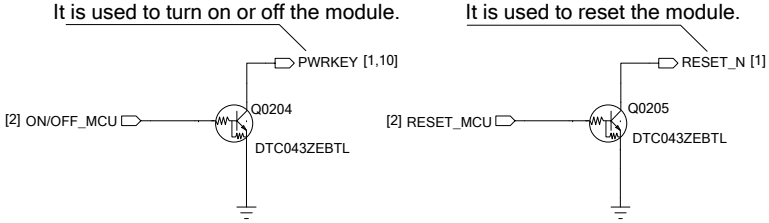
Model	Installed	Not installed
EG9x-E	R0106	R0105, R0107
EG9x-NA	R0105, R0107	R0106

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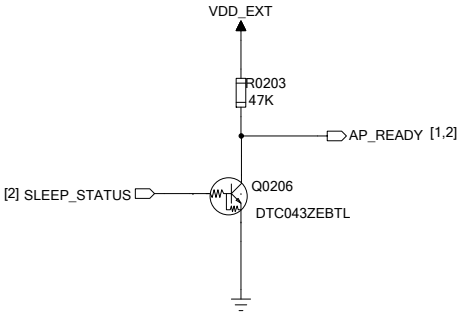
MCU Interface



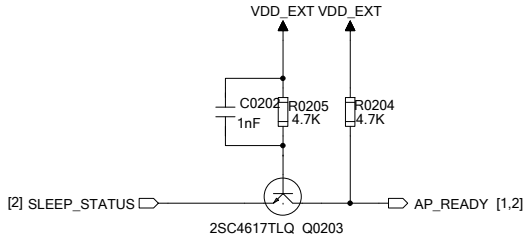




The Necessary Control Circuit



AP_READY High Level Detection



AP_READY Low Level Detection

Notes:

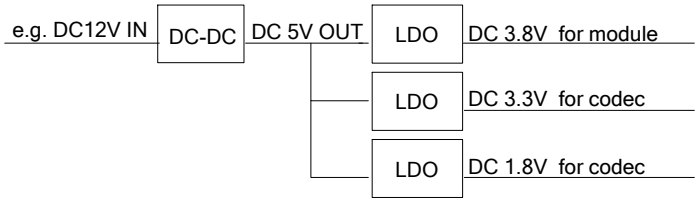
1. U0201 represents customer's MCU.
2. EG9x can only work as a USB device and supports FS/HS modes. To communicate with USB interface, MCU needs to support USB host or OTG function.
   The VBUS pins of MCU and EG9x should be provided by 5V power system for USB detection, and VBUS_CONTROL is used to turn on and off VBUS power supply.
   VBUS_CONTROL is active high.
3. AP_READY is used to detect the MCU's sleep state. For more details about AP_READY, please refer to EG91 and EG95 Hardware Designs and AT Command manuals.
4. Transistor circuits (Q0203～Q0206) are used for level translation.

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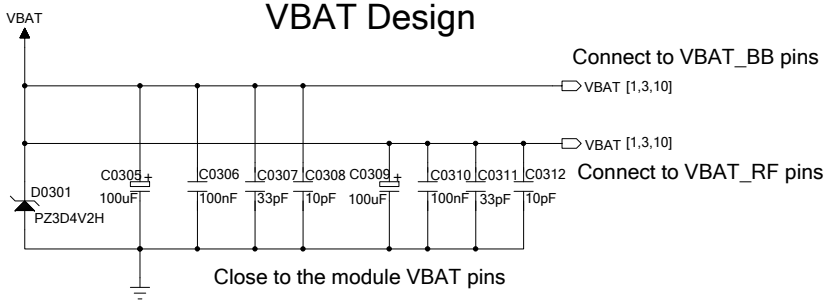
Power Supply Design

DC-DC Application

It is used when the input voltage is above 7V. Use a DC-DC convertor to convert a high input voltage to a 5V output, and then the LDOs will generate a 3.8V typical voltage for the module and 1.8V & 3.3V typical voltages for codec.



VBAT Design

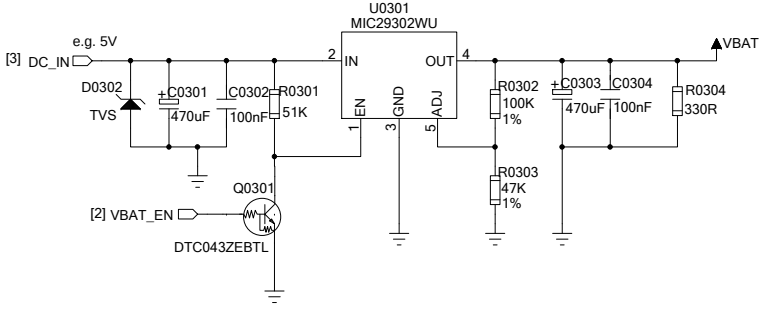


Notes:

1. The power supply must be able to provide sufficient current up to 2A or more.
2. VBAT should be routed in star mode to VBAT_BB and VBAT_RF pins.
3. The recommended operating voltage of VBAT is 3.3V～4.3V.

LDO Application

It is used when the input voltage is below 7V.

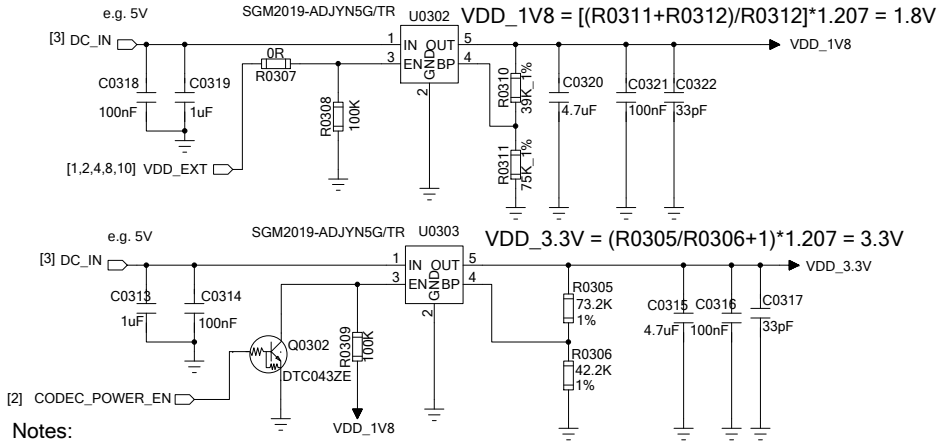


VBAT = (R0302/R0303+1)*1.24 = 3.88V

Notes:

1. The load current is recommended to be greater than 10mA.
2. The power supply must be able to provide sufficient current up to 2A or more.

Supply Power for PCM Codec



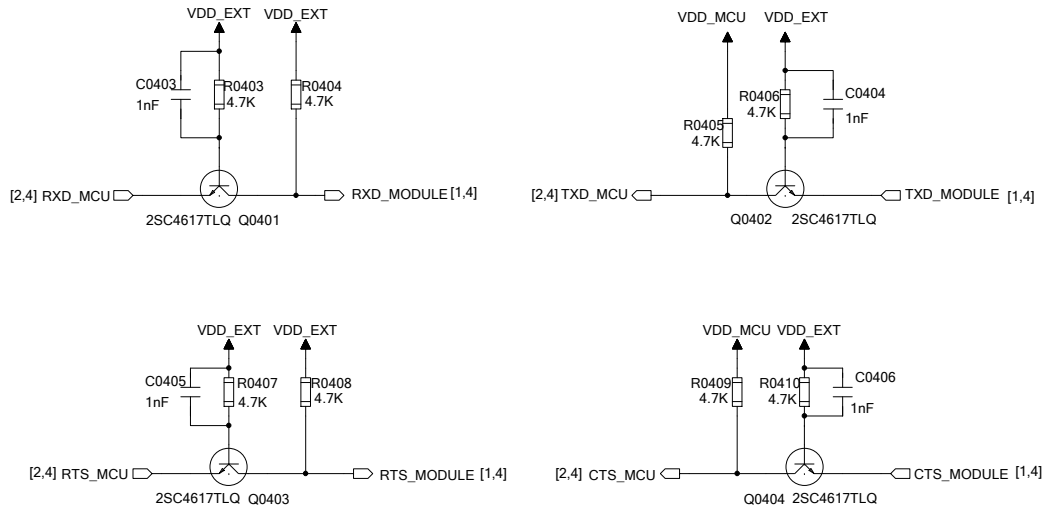
Notes:

1. CODEC_POWER_EN must be at low level in order to ensure the normal output voltage of VDD_3.3V.
   If VDD_3.3V power supply needs to be switched off, please keep CODEC_POWER_EN at high level.
2. The following power-on/off sequences should be complied with to ensure the audio codec works normally.
   Power-on Sequence: power on VDD_1V8 first, then VDD_3.3V.
   Power-off Sequence: power off VDD_3.3V first, then VDD_1V8.

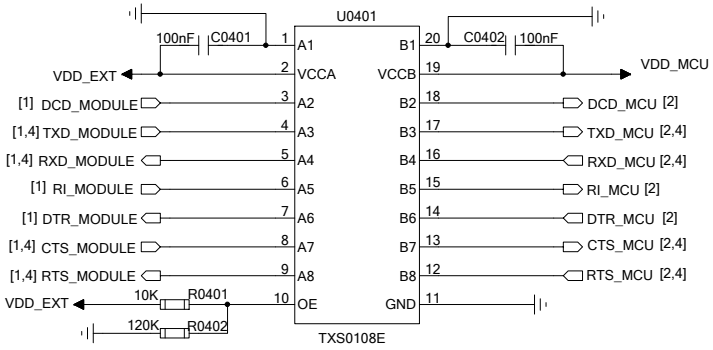
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UART Design

UART Translation - Transistor Solution (Recommended)



UART Translation - IC Solution

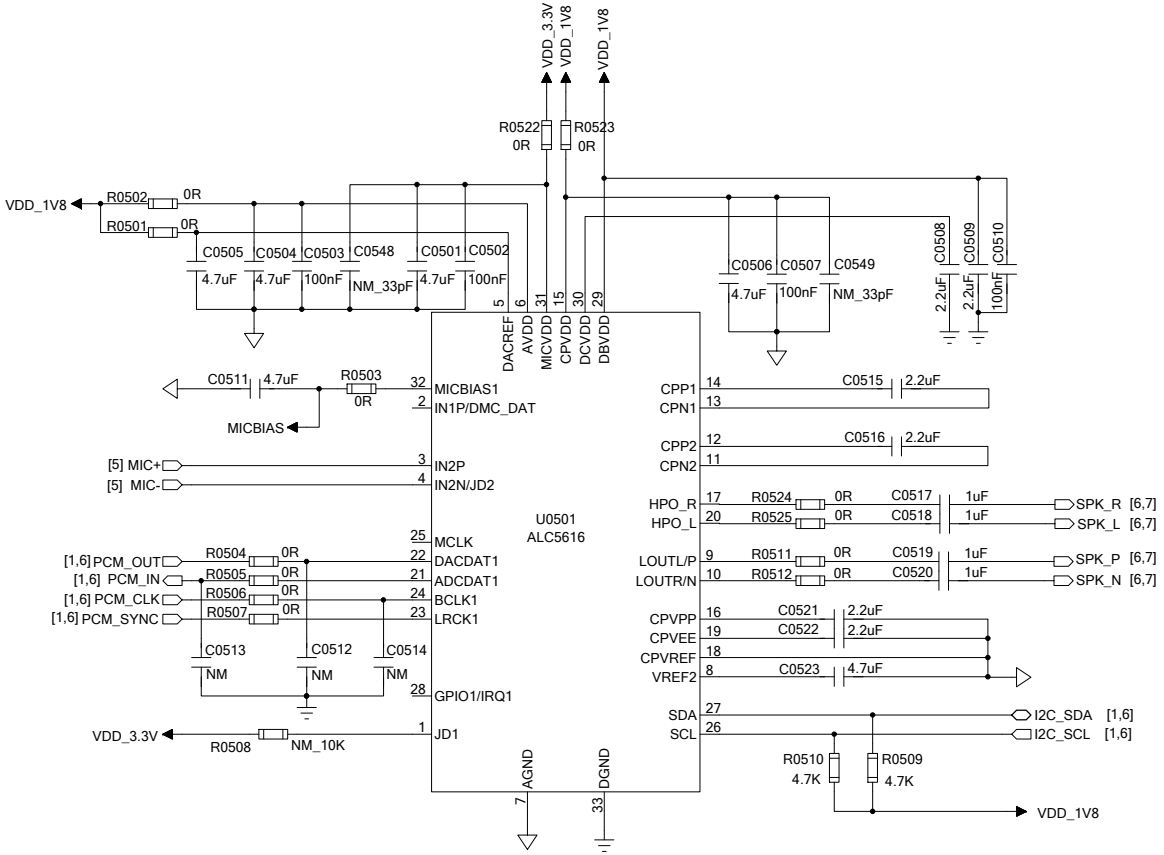


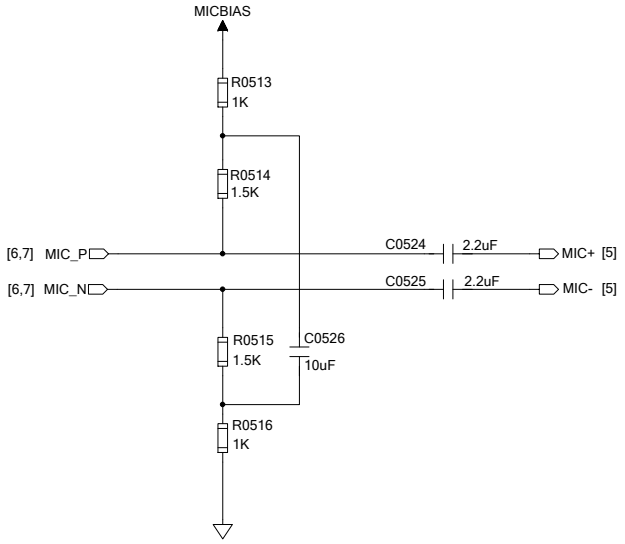
Notes:

1. The power supply voltage of VCCA should not exceed that of VCCB. For more information about TXS0108E, please refer to the datasheet from TI.
2. If a high baud rate is needed, it is highly recommended to install a 1nF capacitor (C0403/C0404) on transistor circiut.
3. The transistor circuit of DTR is similar to that of RTS interface, and the transistor circuit of RI and DCD is similar to that of CTS interface.

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Audio Codec Design (ALC5616)



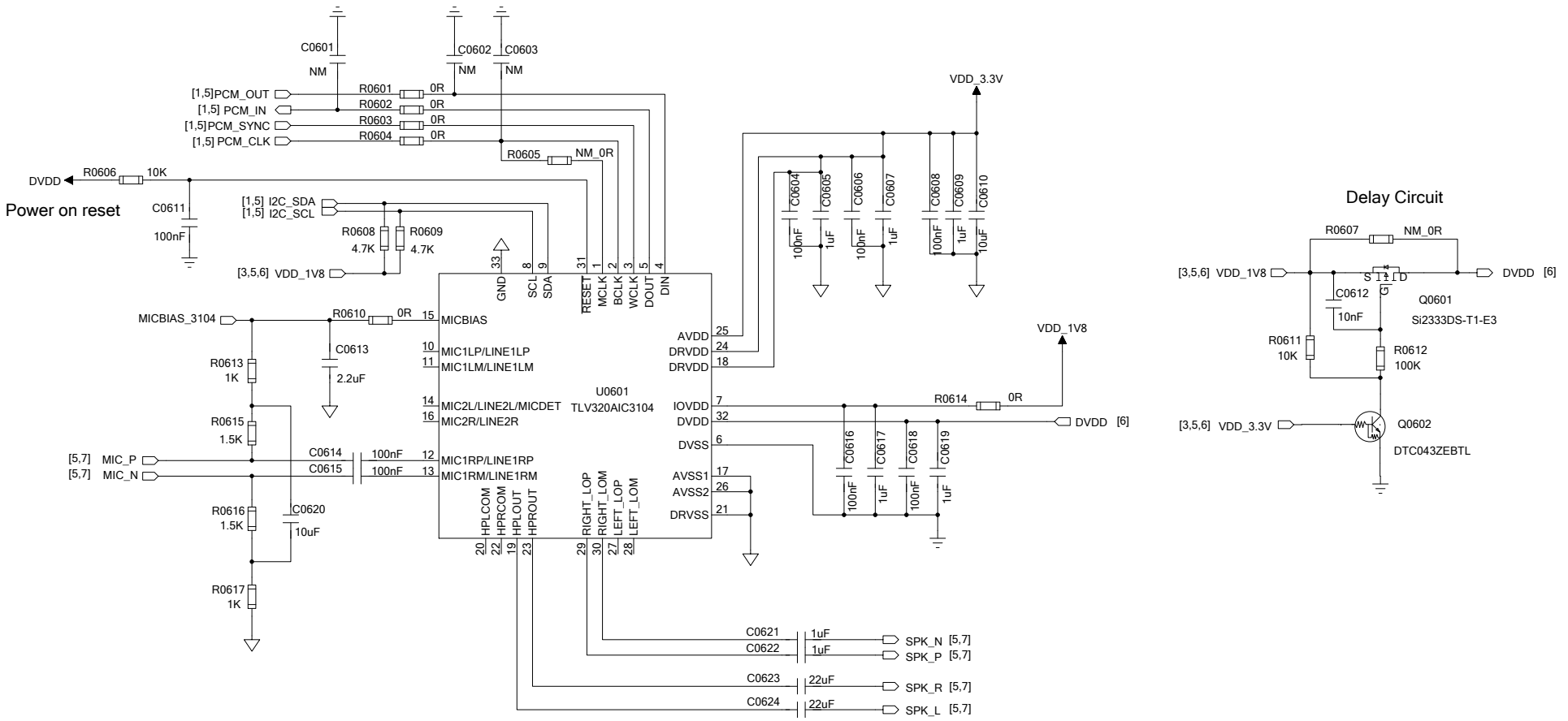


Notes:

1. ALC5616 power-on sequence: DBVDD/I2C pull-up power/AVDD/DACREF/CPVDD -> MICVDD -> software initialization.
2. ALC5616 power-off sequence: close codec function by software -> MICVDD -> DBVDD/I2C pull-up power/AVDD/DACREF/CPVDD.
3. EG9x module will automatically initialize the codec via I2C interface after it is turned on successfully, so all power supplies for the codec need to be powered on before that.

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Audio Codec Design (TLV320AIC3104)



Delay Circuit

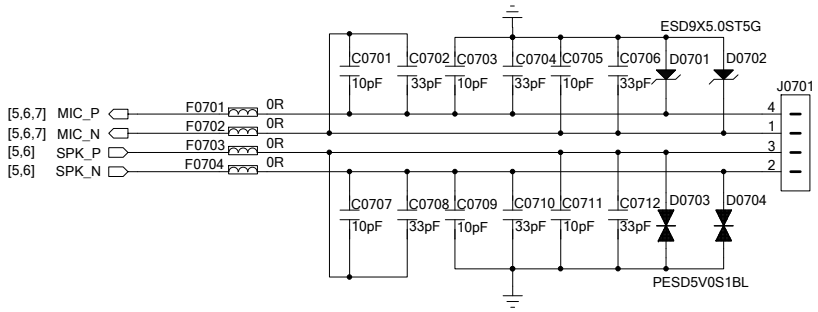
Notes:

1. TLV320AIC3104 power-on sequence: IOVDD -> AVDD/DRVDD -> DVDD -> software initialization.
2. The RC delay circuit, which is assembled with C0612 and R0612, is used to enure that the power on time difference between AVDD and DVDD is within 5ms.
3. The RESET pin must be driven at low level for at least 10ns after all power supplies for TLV320AIC3104 are at their specified values.
4. EG9x module will automatically initialize the codec via I2C interface after it is turned on successfully, so all power supplies for the codec need to be powered on before that.

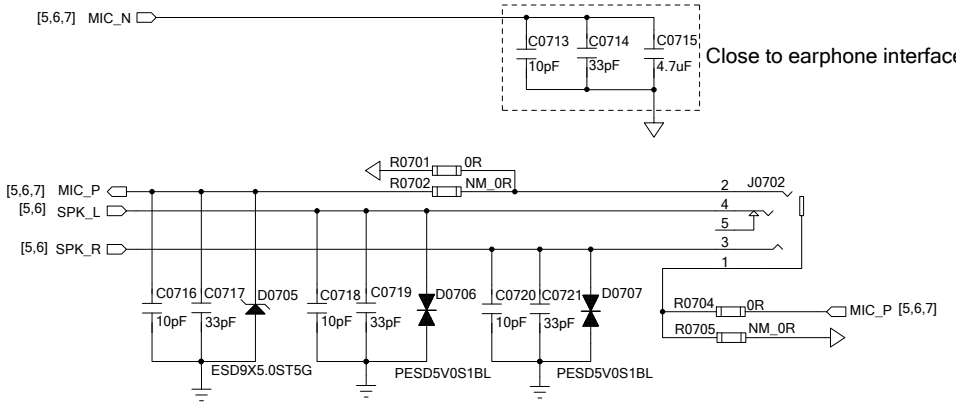
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Audio Interface

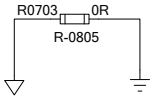
Handset Application



Earphone Application



	CTIA	OMTP
R0702/R0705	NM	M
R0701/R0704	M	NM



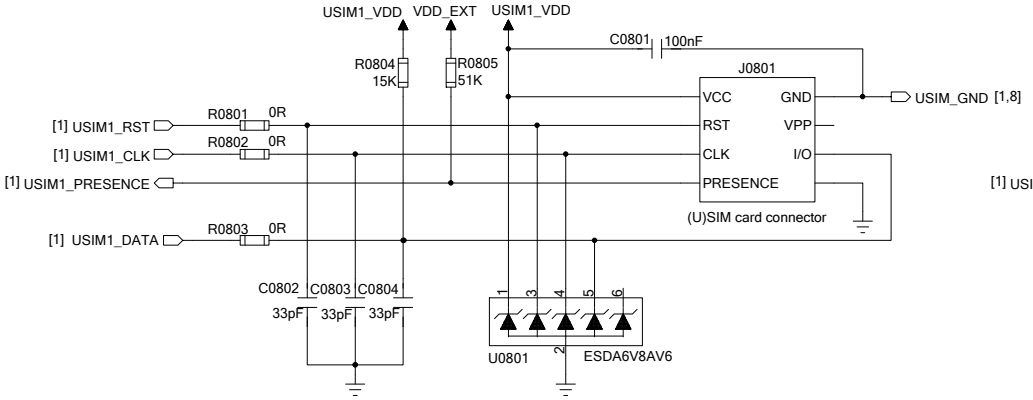
Notes:

1. The analog output only drives earphone and headset. For larger power loads such as speakers, an audio power amplifier should be added in the design.
2. In handset application, route the MIC and SPK signal traces as differential pairs respectively.
3. In earphone application, route the MIC signal traces as differential pairs.
4. All MIC and SPK signal traces should be routed with total grounding and far away from noise such as clock and DC-DC signals, etc.
5. ALC5616 and TLV320AIC3104 cannot be used simultaneously in audio codec

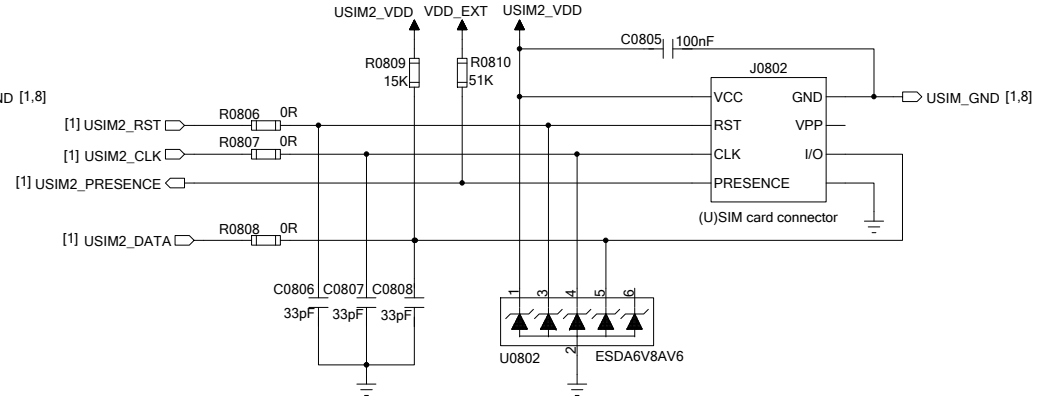
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(U)SIM Interface

USIM1 Interface



USIM2 Interface



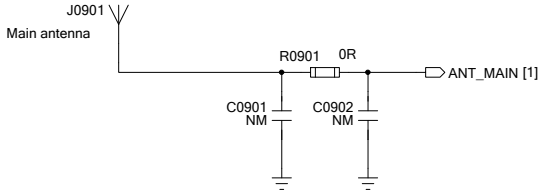
Notes:

1. R0801～R0803 and R0806～R0808 are used for debugging, C0802～C0804 and C0806～C0808 are used for filtering interference of GSM900MHz.
2. It is recommended to connect the (U)SIM card connector GND to the module USIM_GND. If the ground is complete on customers' PCB, USIM_GND can be connected to PCB ground directly.
3. U0801 and U0802 is recommended to be used to offer good ESD protection, and the parasitic capacitance should not be more than 15pF.
4. R0804 and R0809 can improve anti-jamming capability of the (U)SIM circuit, and they should be placed close to the (U)SIM card connector.
5. The decouple capacitor of USIM1_VDD and USIM2_VDD should be less than 1uF and must be placed near to (U)SIM card connector.
6. EG9x module provides two input pins (USIM1_PRESENCE and USIM2_PRESENCE) to detect whether the USIM1 and USIM2 cards are present or not. It supports both low level and high level detections. For more details, please refer to EG91 and EG95 Hardware Designs.

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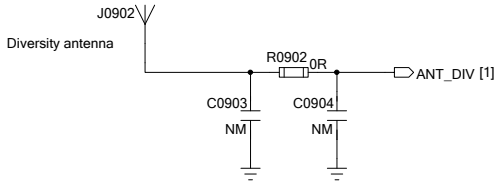
RF and GNSS Designs

Main Antenna Interface



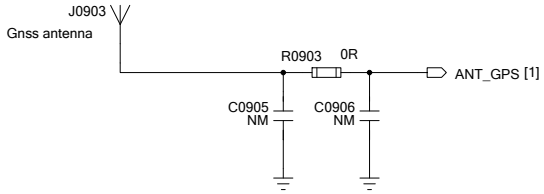
C0901 and C0902 are reserved for impedance matching.

Diversity Antenna Interface



C0903 and C0904 are reserved for impedance matching.

GNSS Antenna Interface

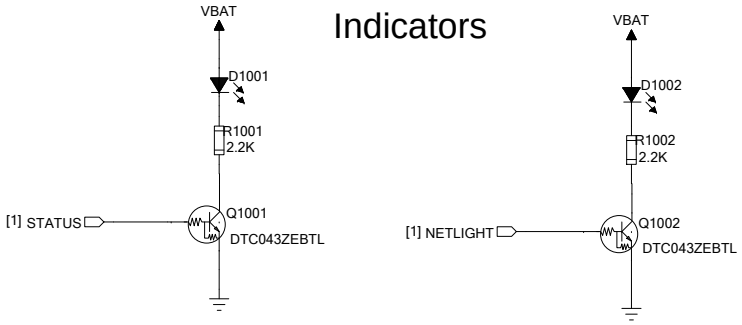


C0905 and C0906 are reserved for impedance matching.

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Indicator and Test Points

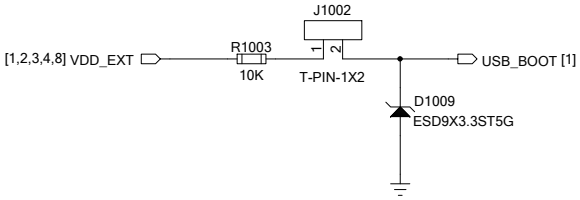
Indicators



Note:

Refer to the document EG91 and EG95 Hardware Designs for more details about NETLIGHT.

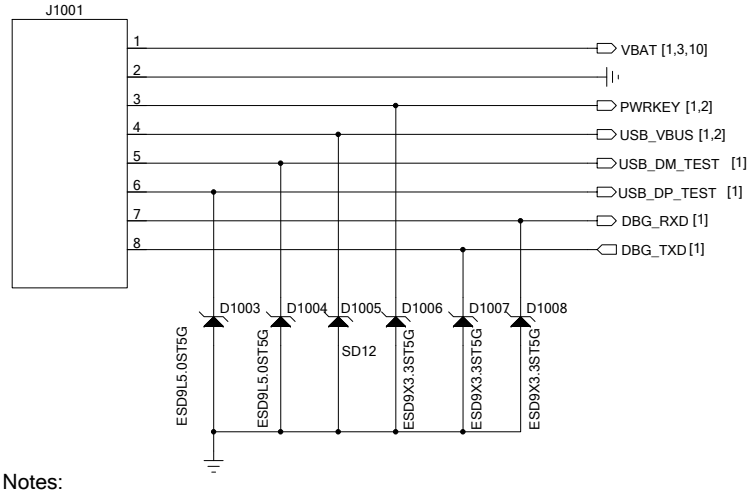
Emergency Download Mode



Note:

When USB_BOOT is at high level, the module will be forced to enter into download mode.

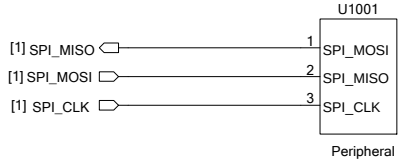
Reserved Test Points



Notes:

1. Both USB and debug UART interfaces are reserved for software debugging.
2. USB interface can also be used to upgrade firmware.
3. Keep USB test points to USB pins as close as possible.
4. Please note that junction capacitance of ESD protection components on USB data lines might influence the signal. Typically, the capacitance should be less than 1pF.

SPI Connection



Note:

It is a dedicated one-to-one connection and no chip select.

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